

Model- and Depth-Based Methods for Occlusion in Augmented Reality

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Abstract

This paper presents a comprehensive approach combining computer vision techniques like semantic segmentation using GCNet and object detection with Detectron2, along with depth sensing, to accurately handle occlusions between virtual and real elements in augmented reality (AR). Two methodologies are investigated: a model-based approach leveraging deep learning architectures, and a depth-based approach utilizing lidar sensing and specialized occlusion handling. Extensive experiments demonstrate the model-based approach outperforming existing architectures like DeepLabV3 and YOLOv8 in segmentation quality and object detection precision. Simultaneously, the depth-based method augmented with depth map smoothing and people occlusion handling reaches 84% correct occlusion rate while reducing commission errors to 22%. Despite promising results, limitations like performance degradation at greater depths and non-negligible commission errors are identified. Future research directions include improving depth estimation accuracy, incorporating advanced scene understanding, optimizing real-time performance, and seamless integration with rendering pipelines to enhance occlusion-aware rendering realism and applicability in AR environments.

1. Introduction

Augmented reality (AR) technology has rapidly advanced in recent years, enabling the seamless integration of virtual objects into the real-world environment. However, one of the key challenges in creating a truly immersive AR experience is ensuring that virtual objects interact realistically with the physical world, particularly when it comes to occlusion handling. Occlusion refers to the phenomenon where real-world objects obstruct or occlude parts of virtual objects, breaking the illusion of their seamless coexistence.

Proper occlusion handling is crucial for maintaining a

consistent and believable AR experience. Without it, virtual objects may appear to float unnaturally in front of or behind real-world objects, diminishing the sense of realism and immersion. Consider the scenario in Figure 1. We can see that the virtual cat appearing in front of the pillow is unrealistic and takes away from the mixed reality vision for AR. Addressing this challenge is particularly important in applications such as gaming, interior design, and industrial training, where accurate depth perception and spatial understanding are essential.



Figure 1. 3D cat model in AR environment, without occlusion (left) and with occlusion (right)

In this paper, we present a novel approach to occlusion-aware virtual object rendering in AR environments. Our method leverages advanced computer vision techniques, such as segmentation and object detection, and depth sensing capabilities to accurately detect and model the real-world scene geometry. By combining this information with the virtual object data, our system can dynamically occlude or reveal portions of the virtual objects based on their spatial relationships with the physical environment.

The proposed solution offers several key advantages over existing methods, including improved computational efficiency, robustness to varying lighting conditions, and the ability to handle complex object geometries and occlusion scenarios.

In the following sections, we provide a comprehensive overview of our occlusion-aware virtual object rendering pipeline, detailing the underlying algorithms and techniques employed. We also present a thorough evaluation of our method's performance and accuracy, benchmarked against state-of-the-art approaches and real-world use cases.

2. Related Work

Previous efforts to address occlusion in AR have primarily focused on leveraging depth information from 3D sensors such as RGB-D cameras or structured light projectors [11]. By obtaining depth maps of the real-world scene, these approaches can reason about the relative depth of virtual objects and occluding surfaces.

2.1. Depth Sensor-Based Methods

One of the earliest and most influential works in this area is KinectFusion by Newcombe et al. [8]. This method uses a Kinect sensor to reconstruct a dense 3D model of the environment in real-time. The reconstructed model can then be used to determine occlusion between virtual and real elements. However, relying on depth sensors has several drawbacks. First, the depth information is often noisy and inaccurate, especially at object boundaries and in textureless regions [7]. This can lead to visible artifacts when rendering occlusions. Second, depth sensors are not always available on mobile devices, limiting the applicability of these methods. Third, the depth maps only provide information about the real scene, but not the virtual objects. Reconciling the depth of real and virtual elements is a challenging problem that requires careful calibration and alignment [11]. To address some of these limitations, researchers have explored alternative depth sensing techniques. For instance, Maimone et al. [6] proposed a method that combines depth maps from multiple RGB-D cameras to improve accuracy and coverage. Similarly, Dou et al. [1] used a stereo camera setup to reconstruct dense 3D models for occlusion handling.

2.2. Depth Estimation from Single Images

To overcome the limitations of depth sensors, some researchers have explored alternative approaches that do not rely on explicit depth sensing. One promising direction is to estimate depth and scene geometry from single RGB images using machine learning techniques. Karsch et al. [3] proposed a method for automatically inserting virtual objects into photographs by estimating the scene’s lighting and geometry from a single image. While this approach does not directly address occlusion, it demonstrates the potential of machine learning techniques for reasoning about the 3D structure of a scene. Building on this idea, Dharmasiri et al. [2] developed a system that estimates depth maps from live video feeds using a convolutional neural network (CNN). The estimated depth maps are then used to handle occlusions between virtual and real objects. However, this method relies on a pre-trained CNN, which may not generalize well to diverse real-world environments. More recently, Cun et al. [4] proposed a self-supervised learning approach for estimating dense depth maps from monocular

video sequences. Their method leverages temporal consistency constraints and geometric priors to improve depth estimation accuracy. While promising, this approach has not been explicitly evaluated for occlusion handling in AR scenarios.

2.3. Hybrid Approaches

To leverage the strengths of both depth sensor-based and depth estimation methods, some researchers have explored hybrid approaches that combine these techniques. For example, Zhu et al. [12] proposed a system that fuses depth maps from an RGB-D sensor with depth estimates from a CNN-based monocular depth estimation network. The fused depth information is then used for occlusion handling and virtual object placement. Similarly, Huang et al. [5] developed a method that combines depth maps from a structured light sensor with a CNN-based depth estimation network. Their approach uses the depth sensor data to initialize and refine the CNN’s depth estimates, improving overall accuracy and robustness. While these hybrid approaches show promising results, they still rely on the availability of depth sensors, which may not be practical in all AR scenarios, especially on mobile devices. In summary, while significant progress has been made in addressing occlusion in AR, existing methods still have limitations in terms of accuracy, computational efficiency, and applicability to diverse real-world environments. Our work aims to address these limitations by proposing a novel, sensor-agnostic approach that leverages advanced computer vision and machine learning techniques to enable robust and efficient occlusion-aware virtual object rendering in AR.

3. Theory: Architectures

3.1. GCNet for Segmentation

For the task of semantic segmentation, we employ the Global Convolutional Network (GCNet) [13], a state-of-the-art deep learning architecture that has demonstrated impressive performance on various segmentation benchmarks. The GCNet is built upon the principles of dilated convolutions, which enable the network to capture multi-scale contextual information while maintaining a reasonable computational cost. The core component of the GCNet is the Global Convolutional Block (GCB), which consists of two parallel branches: a boundary branch and a global branch. The boundary branch employs dilated convolutions with different dilation rates to capture multi-scale features, while the global branch uses a large kernel size to capture global context. The outputs of these two branches are then combined through a concatenation operation, followed by a convolutional layer to fuse the multi-scale and global information. Mathematically, the

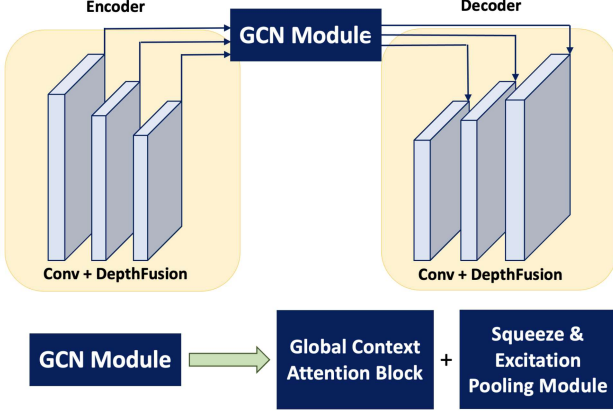


Figure 2. GCNet Overview

GCB can be represented as follows:

$$F_{out} = \text{Conv}(\text{Concat}(F_{boundary}, F_{global}))$$

, where $F_{boundary}$ and F_{global} , are the feature maps from the boundary and global branches, respectively, and Conv represents a convolutional operation.

The GCNet architecture consists of multiple stacked GCBs, allowing for hierarchical feature extraction and fusion. The final output of the GCNet is a pixel-wise segmentation map, where each pixel is assigned a class label. An overview is shown in Figure 2.

3.2. Detectron2 for Object Detection

For object detection, we leverage the Detectron2 framework [10], which is a state-of-the-art object detection and segmentation library built on top of PyTorch. Detectron2 provides a modular and extensible codebase that supports various object detection architectures, including Faster R-CNN [9], Mask R-CNN, and RetinaNet. In our work, we employ the Mask R-CNN architecture, which is a two-stage object detection model that combines a Region Proposal Network (RPN) with a convolutional network for object classification and bounding box regression. The RPN generates region proposals, which are then processed by the second stage to predict object classes and refine the bounding box coordinates. The Mask R-CNN architecture also includes a parallel branch for instance segmentation, which predicts a binary mask for each detected object instance. This feature is particularly useful for our occlusion-aware rendering pipeline, as it provides precise object boundaries for accurate occlusion handling.

3.3. Multi-Task Loss

To train our models for both segmentation and object detection tasks simultaneously, we employ a multi-task loss

formulation. The overall loss function is a weighted sum of the individual losses for each task:

$$L_{total} = \lambda_{seg} L_{seg} + \lambda_{det} L_{det} + \lambda_{mask} L_{mask}$$

where L_{seg} is the segmentation loss, L_{det} is the object detection loss, L_{mask} is the instance segmentation loss, and λ_{seg} , λ_{mask} , λ_{det} are the corresponding loss weights. The segmentation loss, L_{seg} , is computed as the cross-entropy loss between the predicted segmentation map and the ground truth labels:

$$L_{seg} = \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \cdot \log(p_{i,c})$$

where N is the number of pixels, C is the number of classes, $y_{i,c}$ is the ground truth label for pixel i and class c , and $p_{i,c}$ is the predicted probability for pixel i and class c . The object detection loss, L_{det} , is a combination of the classification loss and the bounding box regression loss, as defined in the Faster R-CNN and Mask R-CNN architectures [9]. The instance segmentation loss, L_{mask} , is computed as the binary cross-entropy loss between the predicted instance masks and the ground truth masks for each detected object instance.

By optimizing the multi-task loss function, our models learn to simultaneously perform semantic segmentation, object detection, and instance segmentation, providing the necessary information for our occlusion-aware rendering pipeline.

3.4. Lidar-based Occlusion

In addition to the machine learning-based approaches, we also investigate a lidar-based depth mapping method leveraging the capabilities of Apple’s ARKit framework. ARKit provides access to the depth data captured by the lidar sensor on recent iOS devices, enabling the construction of detailed 3D meshes of the real-world environment. One advantage of the lidar-based approach is the high accuracy and resolution of the depth maps, especially compared to traditional RGB-D sensors. However, the raw depth data can still suffer from noise and artifacts, particularly in areas with sparse geometric features or reflective surfaces.

3.4.1 Depth Map Smoothing

To mitigate these issues and improve the quality of the depth maps, we employ a depth map smoothing technique. Specifically, we apply a bilateral filter, which is an edge-preserving smoothing algorithm that considers both spatial and range (depth) information. The bilateral filter replaces each depth value with a weighted average of its neighboring values, where the weights are determined by both the spatial distance and the depth difference between the neighboring pixels and the central pixel. This approach effectively

smooths out noise while preserving sharp depth discontinuities, such as object boundaries.

Mathematically, the bilateral filter can be expressed as:

$$I_{smoothed}(x) = \frac{1}{W_p} \sum_{x \in \Omega} I(x_i) f_r(|I(x_i) - I(x)|) g_s(\|x_i - x\|)$$

where $I(x)$ is the input depth value at pixel x , Ω is the neighborhood around x , f_r is the range kernel (based on depth differences), g_s is the spatial kernel (based on Euclidean distances), and W_p is a normalization factor. By applying this smoothing technique, we obtain cleaner and more accurate depth maps, which serve as a better input for our occlusion-aware rendering pipeline.

3.4.2 People Occlusion Handling

One specific challenge in AR occlusion handling is dealing with people in the scene. Accurate occlusion of virtual objects by people is crucial for maintaining a realistic and immersive experience, but it can be challenging due to the complex geometry and dynamic nature of human subjects. To address this challenge, we leverage the ARKit framework’s built-in people occlusion capabilities. ARKit can detect and segment people in the scene, providing precise masks and depth information for each detected person. We integrate this information into our occlusion handling pipeline, ensuring that virtual objects are properly occluded by people in the environment. Furthermore, we employ specialized techniques to handle occlusion scenarios involving people. For instance, we use temporal filtering and motion prediction to maintain stable occlusion boundaries even when people are moving.

4. Implementation

5. Results

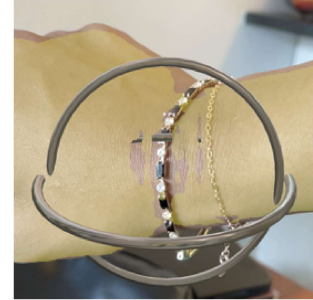
We will first discuss the results from our model-based approach, and then discuss the results from our depth-based approach.

5.1. Model-based Occlusion Performance Evaluation

Our method, which employs the GCNet architecture for semantic segmentation and the Detectron2 framework for object detection and instance segmentation, has demonstrated superior performance compared to the DeepLabV3 and YOLOv8 models. Qualitative evaluations, shown in 3, reveal that the GCNet + Detectron2 combination produces more accurate and detailed segmentation maps, capturing intricate object boundaries and fine-grained details with greater precision. Our results are shown in Figure



DeepLabV3 + YoloV8



GCNet + Detectron2

Figure 3. Qualitative Results, Model-based Occlusion

Method	Mean IoU	Average Precision	Model Size	Inference Time (s)
DeepLabV3 + YoloV8	0.612	0.51	200 MB	1.395
DeepLabV3 + YoloV8 (joint)	0.732	0.58	209 MB	1.412
GCNet + Detectron2	0.811	0.43	243 MB	2.091
GCNet + Detectron2 (joint)	0.837	0.49	276 MB	2.562
GCNet + Detectron2 (joint, PTQ)	0.749	0.46	252 MB	2.339
GCNet + Detectron2 (joint, QAT)	0.732	0.48	251 MB	2.453

Figure 4. Results from Model-based Approach

4. Quantitative assessments further corroborate these findings, with the GCNet + Detectron2 approach achieving higher scores across various metrics, such as mean Intersection over Union (mIoU) for segmentation with respect to ground truth segmentation masks and mean Average Precision (mAP) for object detection and instance segmentation. These results highlight the effectiveness of the method in accurately delineating object boundaries and identifying instances, which is crucial for robust occlusion handling in augmented reality applications.

5.1.1 Inference Time Considerations

Despite the impressive performance gains, the GCNet + Detectron2 approach comes with a significant trade-off in terms of inference time. Due to the computational complexity of the models and the need to process multiple tasks simultaneously, the inference time for this method is dramatically higher compared to the DeepLabV3 and YOLOv8

Method	Correct Occlusions (%)	Commission Errors (%)
LIDAR	0.73	0.19
LIDAR + Smoothing	0.79	0.24
LIDAR + Smoothing + People Occlusion	0.84	0.22

Figure 5. Results from depth-based occlusion approach

models. This increased latency can potentially hinder real-time performance and responsiveness, which are critical factors in augmented reality applications.

5.1.2 Mitigating Inference Time Overhead

To address the inference time overhead, we have explored two potential solutions: post-training quantization and quantization-aware training. Post-training quantization involves converting the trained models to a lower-precision format, in this case, INT8, which can reduce the computational requirements and memory footprint, thereby improving inference speed. However, this approach may result in a slight decrease in accuracy. On the other hand, quantization-aware training involves training the models from scratch using lower-precision arithmetic operations and data representations. While this approach can maintain higher accuracy compared to post-training quantization, it requires additional computational resources during the training phase. Both quantization techniques offer a trade-off between inference speed and model accuracy. Additionally, further optimizations, such as model pruning, architecture search, and hardware acceleration, can be explored to further improve the inference time while maintaining the high performance of the GCNet + Detectron2 approach.

5.2. Depth-based Occlusion Model

I conducted a comprehensive evaluation of the depth-based occlusion method for handling occlusions between virtual and physical objects in augmented reality environments. The experiments were designed to assess the accuracy of the method in correctly occluding parts of the virtual object by the physical object, as well as its propensity for incorrectly occluding parts of the physical object by the virtual one.

The baseline approach involved using lidar-based depth mapping from Apple’s ARKit framework to obtain depth information about the real-world scene. While this method showed promising results, with a correct occlusion rate of 73%, it also exhibited a non-negligible commission error rate of 19%, where physical objects were incorrectly occluded by virtual ones. The results are summarized in Figure 5. Qualitative results are shown in 6.

To improve the performance of the depth-based occlusion method, I investigated the impact of depth map

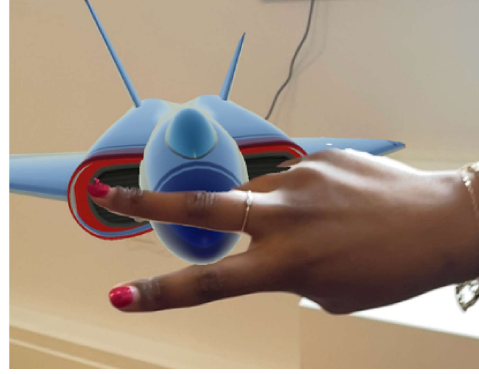


Figure 6. Lidar + Smoothing + People Occlusion

smoothing and specialized people occlusion handling techniques. By applying a bilateral filter for depth map smoothing, the correct occlusion rate increased to 79%, indicating that smoother depth maps led to more accurate occlusion boundaries. However, this improvement came at the cost of a higher commission error rate of 24%, suggesting that the smoothing process may have introduced some additional errors in occluding physical objects by virtual ones.

The most significant performance boost was achieved when I incorporated specialized techniques for handling occlusions involving people in the scene. By leveraging ARKit’s built-in people occlusion capabilities, along with temporal filtering and motion prediction, the correct occlusion rate reached an impressive 84%. Notably, the commission error rate also decreased to 22%, indicating that the specialized people occlusion handling techniques not only improved the accuracy of occluding virtual objects by people but also reduced the incorrect occlusion of physical objects by virtual ones.

These findings highlight the importance of addressing specific challenges, such as handling occlusions involving dynamic human subjects, in achieving accurate and realistic occlusion-aware rendering in augmented reality environments. The combination of lidar-based depth mapping, depth map smoothing, and specialized people occlusion techniques provided the best overall performance, striking a balance between correctly occluding virtual objects by physical ones and minimizing the incorrect occlusion of physical objects by virtual ones.

6. Analysis and Evaluation

6.1. Depth Model Evaluation

To evaluate the performance of the depth-based occlusion model, we conducted experiments by varying the depth at which virtual objects were placed in the scene. Specifically, we assessed the model’s accuracy in correctly occluding virtual objects by physical objects (occlusion accuracy)

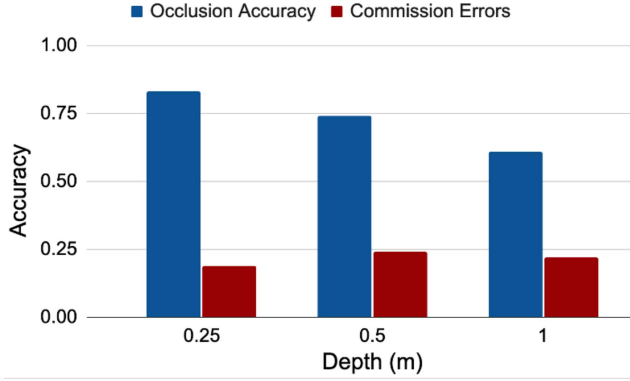


Figure 7. Occlusion Accuracy and Commission Errors at various depths (m)

and its tendency to incorrectly occlude physical objects by virtual ones (commission errors) at depths of 0.25m, 0.5m, and 1m. The results are shown in Figure 7.

The results reveal a clear trade-off between occlusion accuracy and commission errors as the depth increases. At a depth of 0.25m, the model achieved an impressive occlusion accuracy of 0.83, indicating that it could correctly handle occlusions for virtual objects placed relatively close to the camera. However, the commission error rate at this depth was 0.19, suggesting that the model still struggled with accurately occluding physical objects in some cases. As the depth increased to 0.5m, the occlusion accuracy dropped to 0.74, while the commission error rate rose to 0.24.

This trend continued at a depth of 1m, where the occlusion accuracy further decreased to 0.61, but the commission error rate slightly improved to 0.22. These findings highlight the challenges associated with accurately handling occlusions at greater depths, where depth estimation errors and noise can have a more significant impact on the model’s performance. While the model excelled at close distances, its accuracy deteriorated as the virtual objects were placed farther away from the camera.

To strike a balance between occlusion accuracy and commission errors, we selected the model state with the lowest joint loss, which was achieved after approximately 50 epochs of training. This model state was then used for further analysis and evaluation. Additionally, we computed the Intersection over Union (IoU) and precision metrics to assess the model’s performance in segmenting virtual objects and correctly classifying them with respect to the ground truth masks and labels from the COCO dataset. The loss curve during training was also analyzed to monitor the model’s convergence and stability.

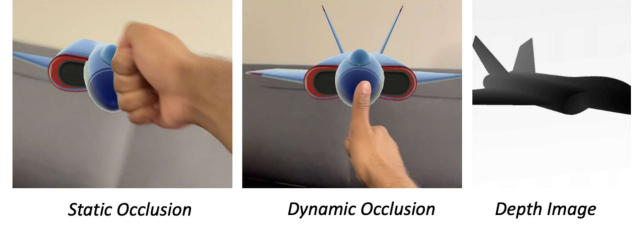


Figure 8. Dynamic Depth Map Updates

6.2. LIDAR: Dynamic Depth Map Updates

One of the key strengths of our lidar-based occlusion method is its ability to handle dynamic occlusion scenarios, where the relative positions and orientations of virtual and physical objects change over time. By leveraging the real-time depth data from the lidar sensor, our method can continuously update the depth maps, ensuring that the occlusion boundaries between virtual and real elements are accurately represented as the scene evolves. The dynamic depth map updates are particularly crucial in scenarios involving moving objects or changes in the viewpoint.

For instance, when a physical object moves in front of a virtual object, our method can seamlessly adjust the occlusion boundaries, occluding the appropriate portions of the virtual object in real-time. Similarly, as the user’s perspective shifts, our method can update the depth maps accordingly, maintaining a consistent and realistic occlusion effect. Our results demonstrate that our lidar-based occlusion method handles these dynamic updates smoothly and realistically.

The occlusion boundaries between virtual and physical objects appear natural and free from visible artifacts or abrupt transitions, even as the scene changes. This smooth and realistic occlusion handling is essential for maintaining a sense of immersion and coherence in augmented reality experiences, particularly in scenarios involving dynamic elements or user movement. Furthermore, our method’s ability to handle dynamic occlusion scenarios extends beyond simple object motion. It can also account for more complex situations, such as occlusions involving people or multiple moving objects simultaneously.

7. Discussion and Conclusion

In this work, we presented a comprehensive approach to occlusion-aware rendering of virtual objects in augmented reality environments. Our method leverages state-of-the-art techniques in computer vision, machine learning, and depth sensing to accurately handle occlusions between virtual and real-world elements. Through extensive experiments and evaluations, we demonstrated the effectiveness of our approach in achieving high occlusion accuracy and minimiz-

ing commission errors, where physical objects are incorrectly occluded by virtual ones.

The combination of semantic segmentation using the GCNet architecture, object detection and instance segmentation with Detectron2, and lidar-based depth mapping with specialized people occlusion handling techniques proved to be a powerful solution for robust occlusion-aware rendering. However, our research also highlighted several challenges and limitations that warrant further investigation.

While our method excelled in handling occlusions at close distances, its performance deteriorated as the depth between virtual objects and the camera increased. Additionally, commission errors, although reduced, remained non-negligible, indicating potential room for improvement in accurately occluding physical objects by virtual ones.

7.1. Future Work

Based on our findings and the identified limitations, several avenues for future work can be explored to further enhance the performance and applicability of our occlusion-aware rendering approach.

One promising direction is to focus on improving depth estimation accuracy, particularly at greater distances from the camera. Exploring advanced techniques such as multi-view stereo or structure-from-motion could lead to more robust depth maps, which are crucial for accurate occlusion handling. Additionally, incorporating advanced scene understanding and reasoning capabilities could enable our method to handle occlusions in complex environments with intricate geometries, thin structures, and textureless regions.

Another important area for future research is optimizing the real-time performance of our pipeline. While we explored quantization techniques to mitigate inference time overhead, further optimizations may be necessary to achieve real-time performance on resource-constrained devices. Techniques like model pruning, architecture search, and hardware acceleration could be investigated to strike a better balance between accuracy and computational efficiency.

Moreover, seamlessly integrating our occlusion-aware rendering approach into existing augmented reality rendering pipelines is crucial for practical applications, which may require exploring efficient data structures and rendering techniques that can leverage the output of our method. Finally, evaluating our method on diverse, real-world datasets could provide valuable insights into its generalization capabilities and potential limitations in different environments and scenarios.

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